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DATA MINING

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BUSINESS INSIGHTS

1. The dataset reveals that universities with higher academic reputation ranks higher in the overall list. This shows how scholarly recognition affects the universities' ranks.
2. The rankings of universities in developing countries have improved in 2025 compared to 2024. This shows that international collaborations greatly impact the education market of a university.
3. Universities with high "International Faculty Ratio" and "International Student Ratio" tend to have higher academic and employer reputations.
4. "Citations per Faculty" has a strong positive correlation with a university's research output level, more than sheer faculty count.
5. Smaller institutions (size M) that achieved top rankings, like MIT, score consistently high across academic, employment, and research criteria. This shows that institutional size does not limit global excellence if performance is well-rounded.
6. Universities that excel in employment outcomes and sustainability tend to achieve higher overall rankings, indicating that a focus on graduate employability and environmental responsibility significantly enhances global standing

DATA UNDERSTANDING

The QS World University Rankings 2025 dataset contains global university rankings, their scores across various criteria, and other data such as country, region, classification size of the institution, etc. This type of data is structured and quantitative. The attributes of the dataset are explained in the table below.

Attributes	Description
Rank	The QS Ranking of universities
Year	The year for the rank of universities, either 2024 or 2025
Institution name	Name of each academic institution
Country	Country of where the university is located at
Region	The continent of which the university is located at
Institution size	Size of the universities (S, M, L, XL)
Status	Status of the universities whether they are public or private
Academic Reputation Score	The score of each university based on global academic reputation survey
Academic Reputation Rank	Rank of each university based on academic reputation
Employer Reputation Score	The score of each university based on global employer reputation survey
Employer Reputation Rank	Rank of each university based on employer reputation
Faculty Student Score	The score of each university based on student-to-faculty ratio
Faculty Student Rank	Rank of each university based on student-to-faculty ratio
Citation per Faculty Score	The score of each university based on research impact by faculties in the university
Citations per Faculty Rank	Rank of each university based on research impact by faculties in the university
International Student Score	The score of each university based on the representation of international diversity of faculties in the university
International Student Rank	Rank of each university based on the representation of international diversity of faculties in the university

International Research Network Score	The score of each university based on the global research collaboration
International Research Network Rank	Rank of each university based on the global research collaboration
Employment Outcomes Score	The score of each university based on graduates' employability and success
Employment Outcomes Rank	Rank of each university based on graduates' employability and success
Sustainability Score	The score of each university based on sustainability initiatives and performance
Sustainability Rank	Rank of each university based on sustainability initiatives and performance
Overall Score	The final score of the universities used to determine the universities' ranking

DATA CHARACTERISTICS

The QS World University Rankings 2025 dataset is packed with organized and quantifiable data that showcases how various international universities are performing. Each entry corresponds to a specific institution and includes rankings based on multiple criteria, along with numerical ratings. Plus, this dataset is great for tracking changes over time since it spans two years: 2024 and 2025.

Key characteristics include;

1. **Temporal Dimension:** By adding rankings for the years 2024 and 2025, this dataset makes it easier to analyze trends over time, like keeping an eye on how the rankings of developing nations are progressing.
2. **Multivariate Structure:** This includes a variety of performance indicators such as the ratios of international faculty and students, job placement rates, citations per faculty member, academic reputation, employer reputation, and sustainability efforts. Together, these elements enable us to understand complex relationships and make multi-faceted judgments.
3. **Normalized and Ranked Metrics:** Most attributes are presented as both raw scores and their corresponding global ranks. This allows for a thorough examination, both in absolute terms and in comparison with other institutions.
4. **Categorical Classifications:** When it comes to classifying and evaluating institutions across various dimensions, attributes like Country, Region, Institution Size, and Status (whether public or private) come in handy.
5. **Correlation Potential:**
 - Citations per faculty strongly correlate with research impact, reflecting high-quality academic output.
 - Universities with strong academic and employer reputations often attract more international faculty and students, contributing to higher internationalization scores.
 - Universities that score well in Employment Outcomes and Sustainability tend to achieve higher overall rankings, reflecting their growing importance.
6. **Size Neutrality:** It turns out that smaller universities, like MIT, still manage to score high on rankings, regardless of their size (whether small, medium, large, or extra-large). This suggests that the quality of performance really trumps the scale of the institution and excel in rankings due to QS's normalized metrics, such as citations per faculty, which prioritize quality over scale.
7. **Geopolitical Trends:** QS 2025 data shows improved rankings for universities in emerging regions, such as Asia and the Middle East, driven by international collaboration. This really emphasizes how important international cooperation initiatives can be.
8. **Targeted Insights:** You can tap into this dataset to pull out valuable corporate intelligence related to internationalization, research priorities, sustainability efforts, and strategic enhancements for universities. This dataset enables stakeholders to extract insights on

internationalization, research priorities, sustainability, and strategic planning for universities.